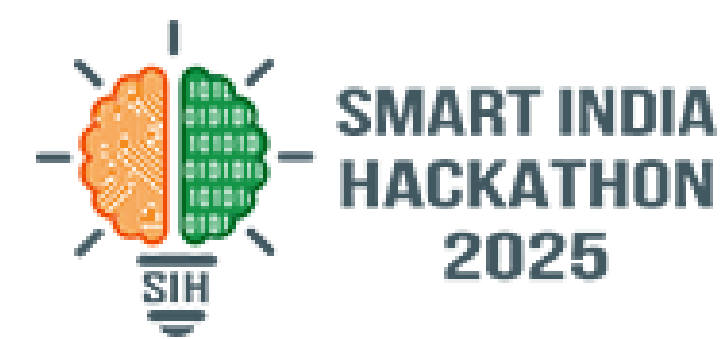
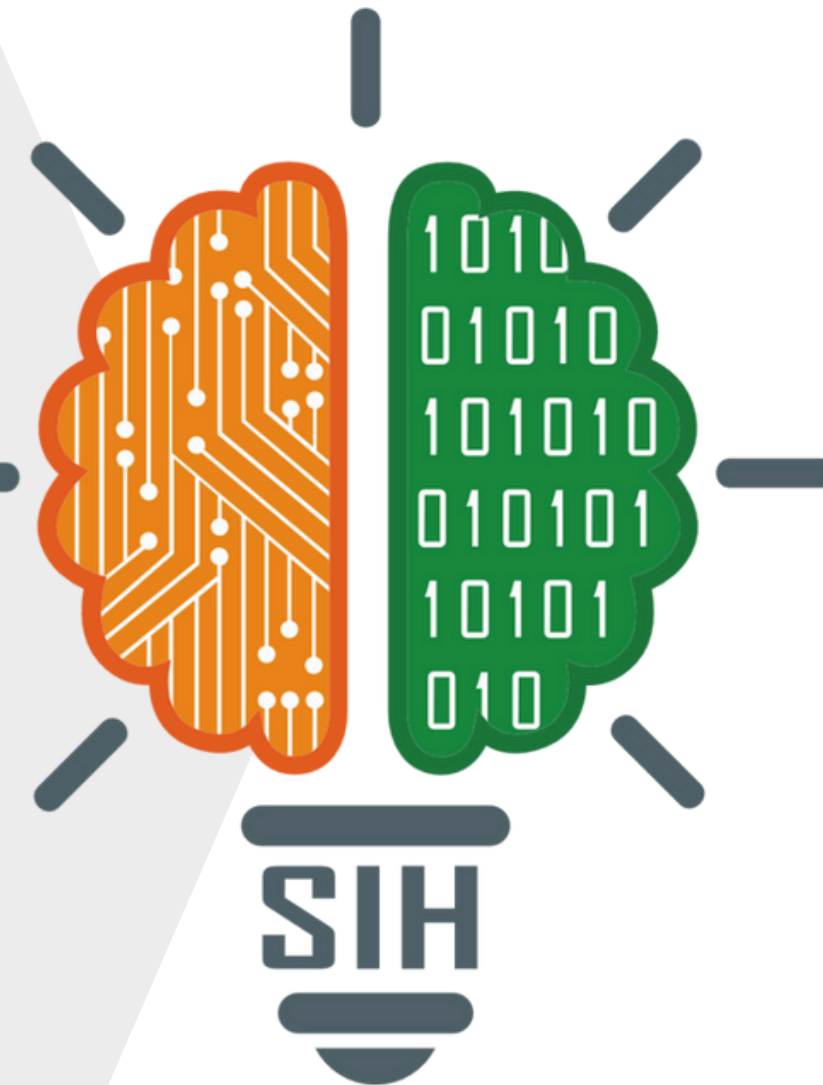


SMART INDIA HACKATHON 2025



DigiShakti

- Problem Statement ID :- SIH 25019
- Problem Statement Title :- Digital Learning platform for Rural Students In Nabha.
- Theme :- Smart Education
- PS Category :- Software
- Team Name :- (VisionX)



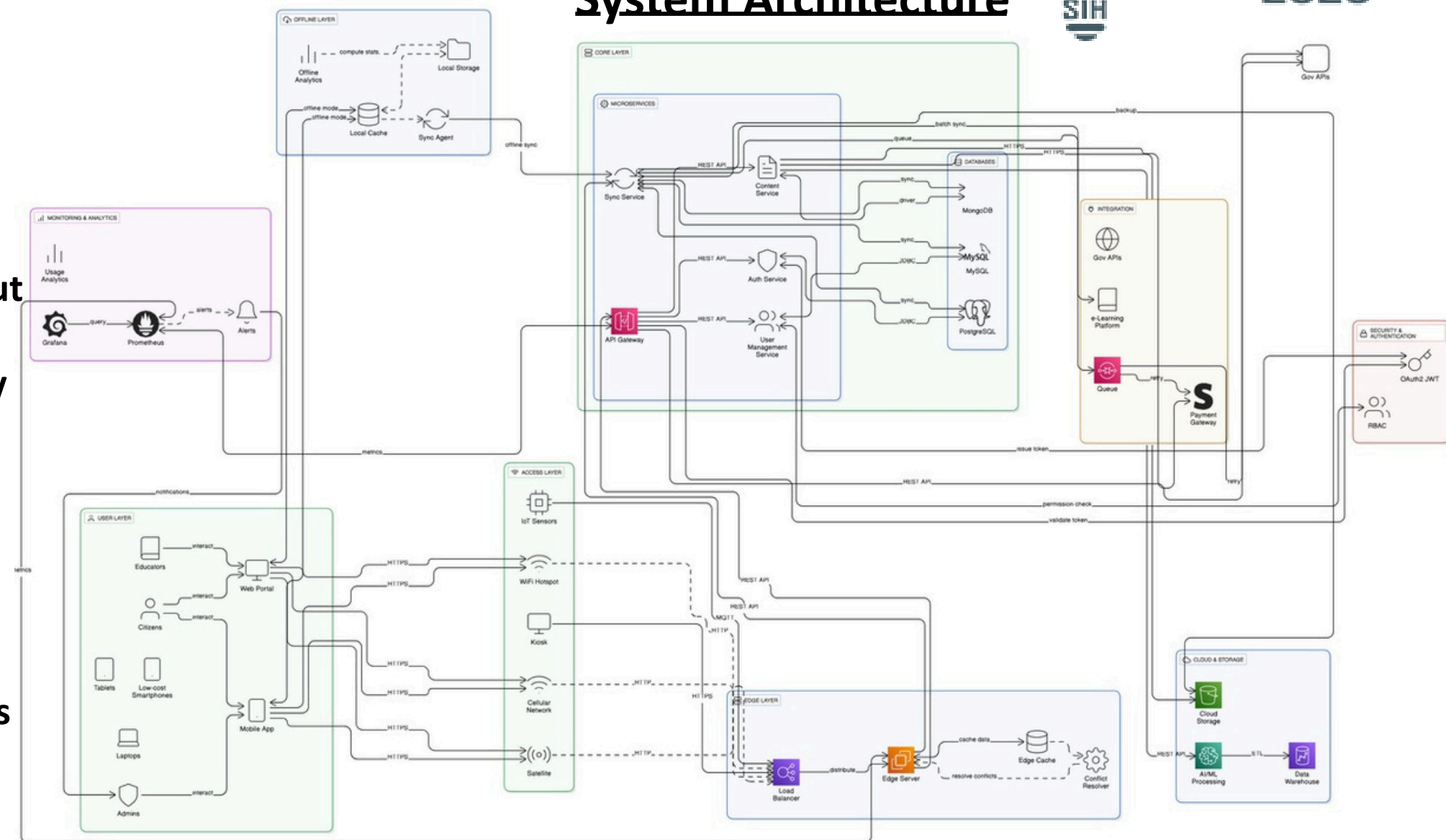
Proposed Solution (DigiShakti)

Problem

- Most rural schools in Punjab don't have proper computer labs or reliable internet.
- ASER 2024 shows only 34% of Class 3 kids can read a Class 2 text, and about 51% can do subtraction.
- Even though more than 85% of rural children now have smartphones, they mostly use them for entertainment, not learning.

Solution

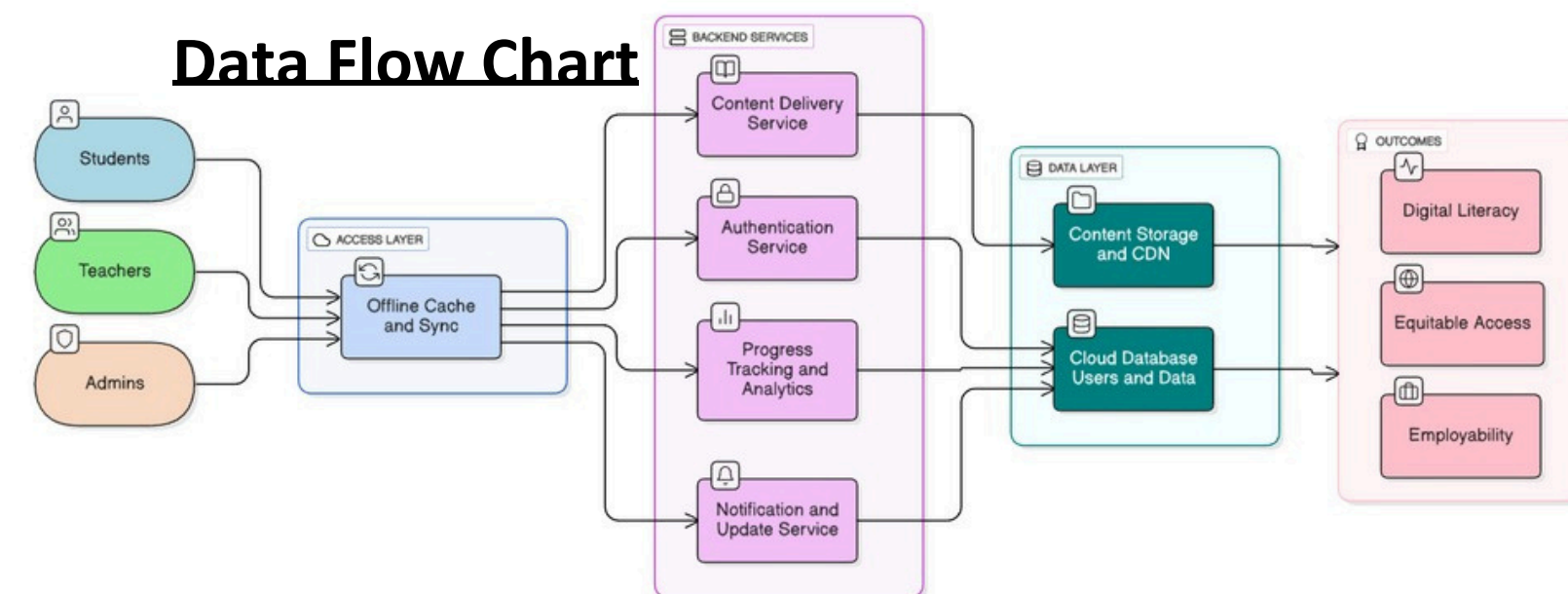
- DigiShakti is a mobile and web app that lets rural students learn even when there's no internet.
- It has simple lessons, videos, and quizzes in Punjabi and Hindi.
- Teachers can see how students are doing through dashboards, which helps them give the right support.
- The app is light, so it runs smoothly on low-end smartphones that most students already have.



Uniqueness of Our System

- Works even without internet – unlike most e-learning apps.
- Combines basic digital literacy + school subjects in one platform.
- Uses the smartphones already in students' hands, so no big cost for new devices.

Data Flow Chart



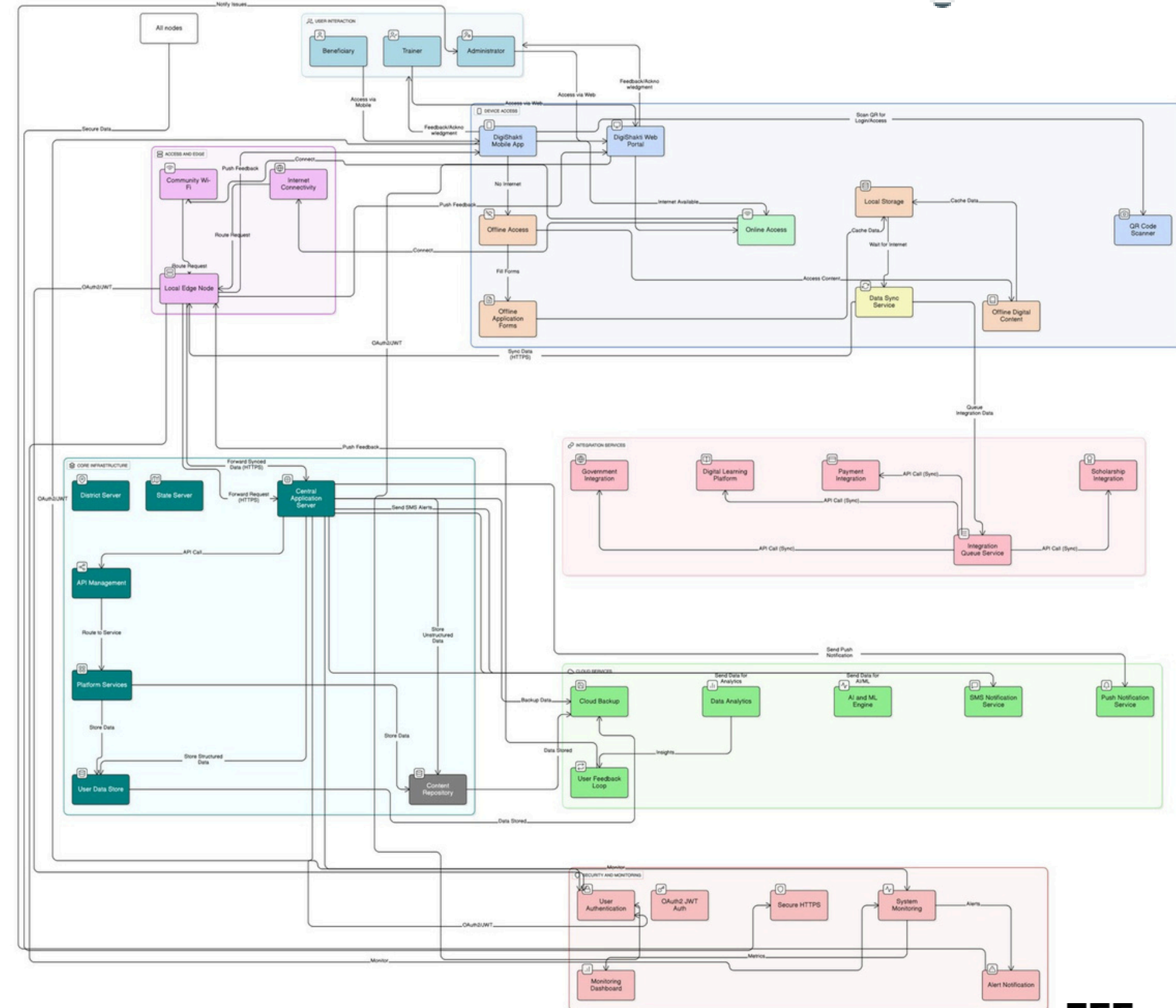
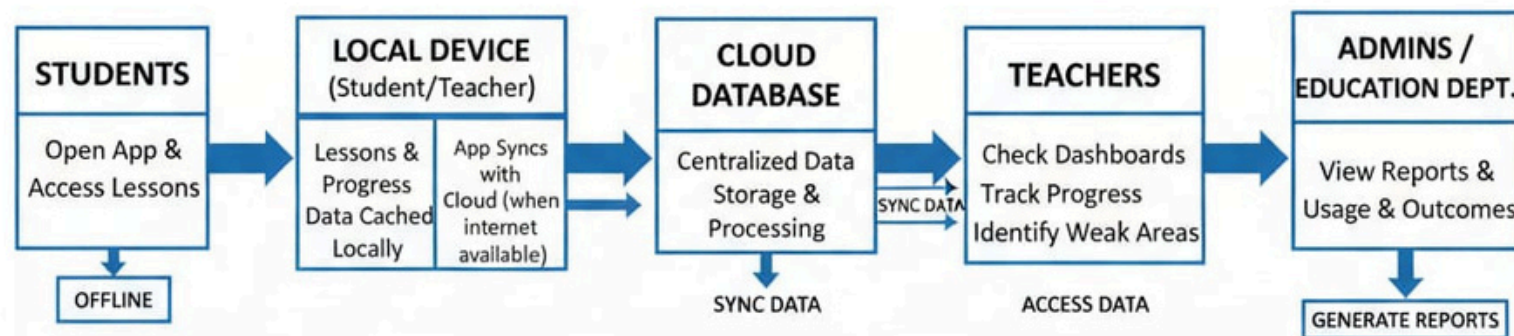
TECHNICAL APPROACH

Tech Stack

Core Stack

- **Mobile App (Android-first):** Kotlin/Java, built lightweight for low-end phones.
- **Web App:** React + Tailwind CSS for teachers and admins.
- **Backend:** Node.js + Express with PostgreSQL/MongoDB for secure and scalable data handling.
- **Offline Mode:** SQLite/Realm for caching lessons, videos, and quiz results locally.
- **Sync Service:** Handles uploads/downloads automatically when the internet is available.
- **Cloud Hosting:** Deployed on AWS/GCP for scalability and data safety.

Workflow



Feasibility.



Policy Alignment

Directly supports NEP 2020 and Digital India Mission, which call for digital literacy, local language content, and inclusive access.



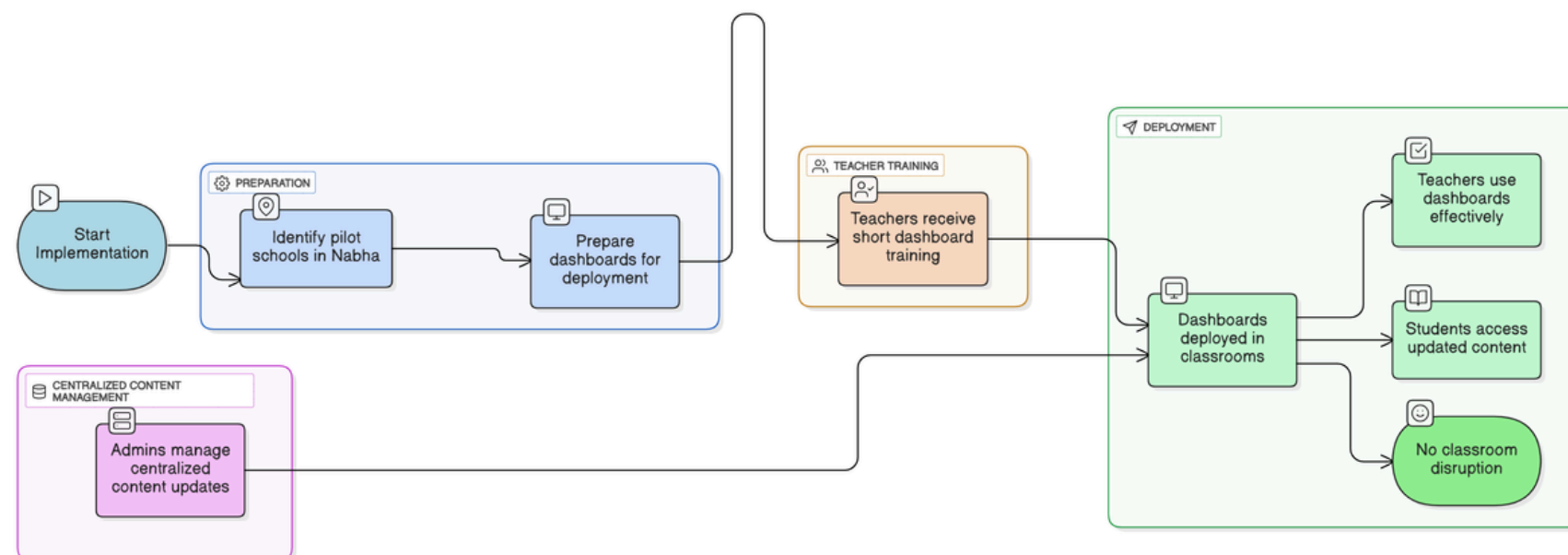
Device Access

- Over 85% of rural children in Punjab have smartphones (Tribune India, 2023). This allows DigiShakti to run without waiting for costly labs or reliable internet.

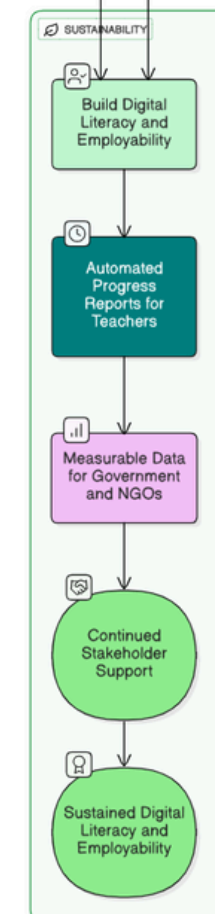
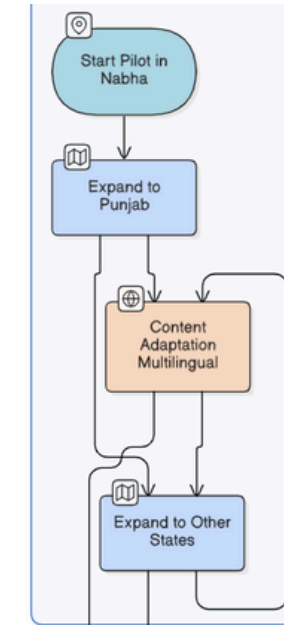
Technology Readiness



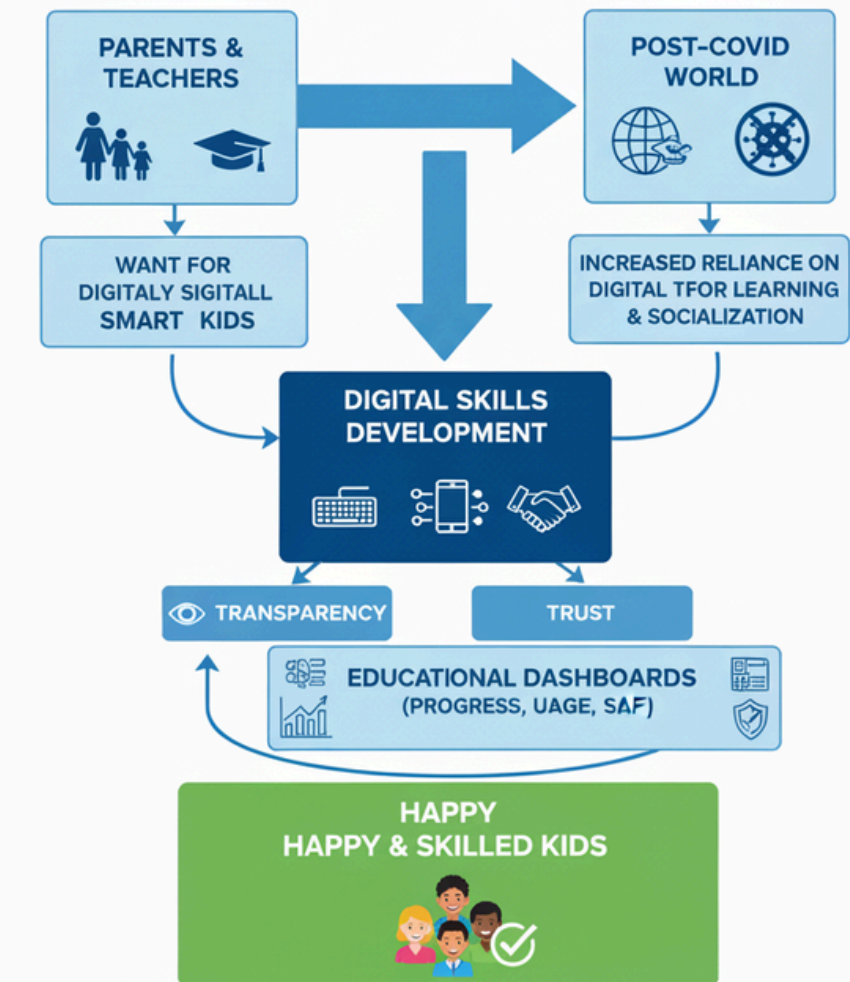
- Offline-first models like iDream Education and Khan Academy Lite already show success in rural India. DigiShakti uses a lightweight, open-source tech stack → reliable, low maintenance, and scalable.



Viability.



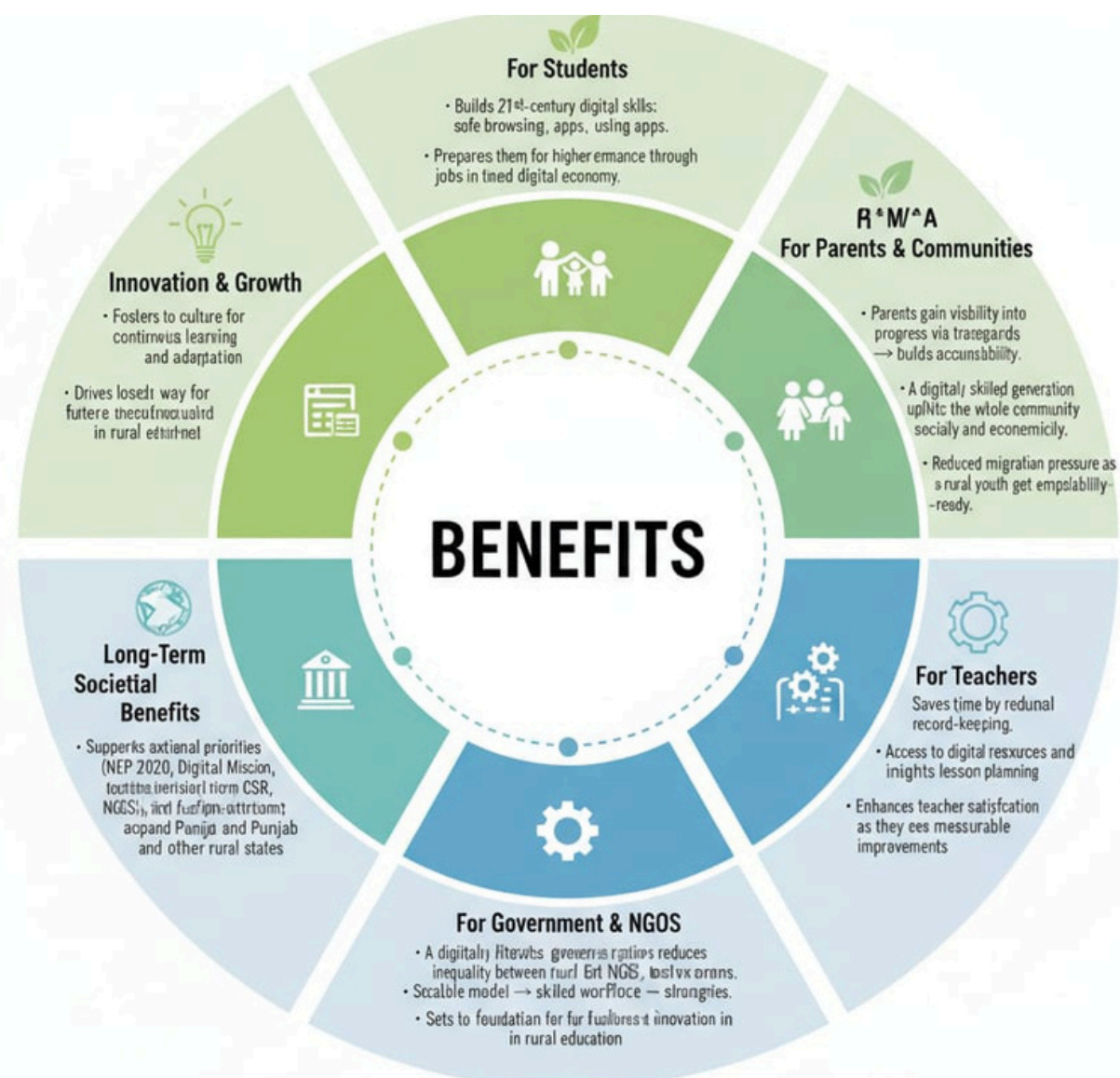
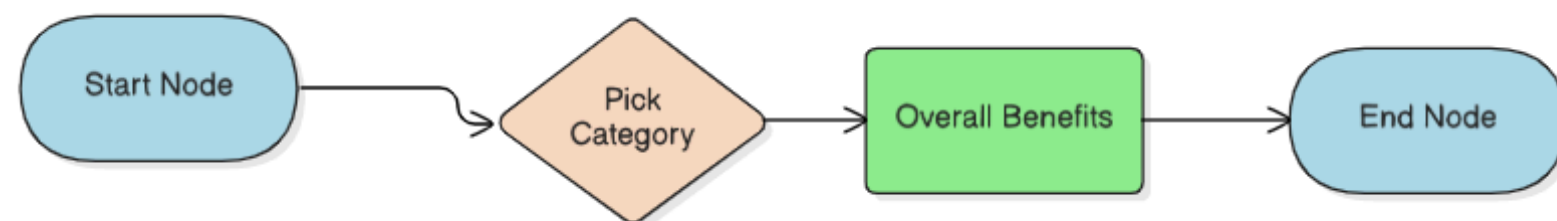
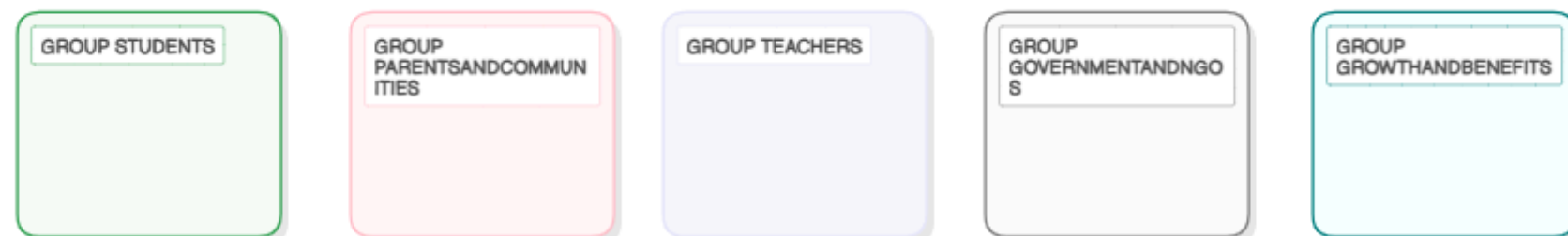
SOCIAL ACCEPTANCE: DIGITAL SKILLS (EASY FLOW)



Financial

- Low-Cost:** The program utilizes existing smartphones, making it highly affordable.
- Sustainable Funding:** It is financed through a mix of CSR, Punjab Government schemes, and NGOs.
- Broad Reach:** The program is scalable and inclusive, allowing it to reach many students.

IMPACT AND BENEFITS



Impact

Digital Inclusion

Gives rural students access to structured digital learning without depending on constant internet.

Narrows the rural–urban education gap by making modern resources available in local languages (Punjabi/Hindi).

Academic Improvement

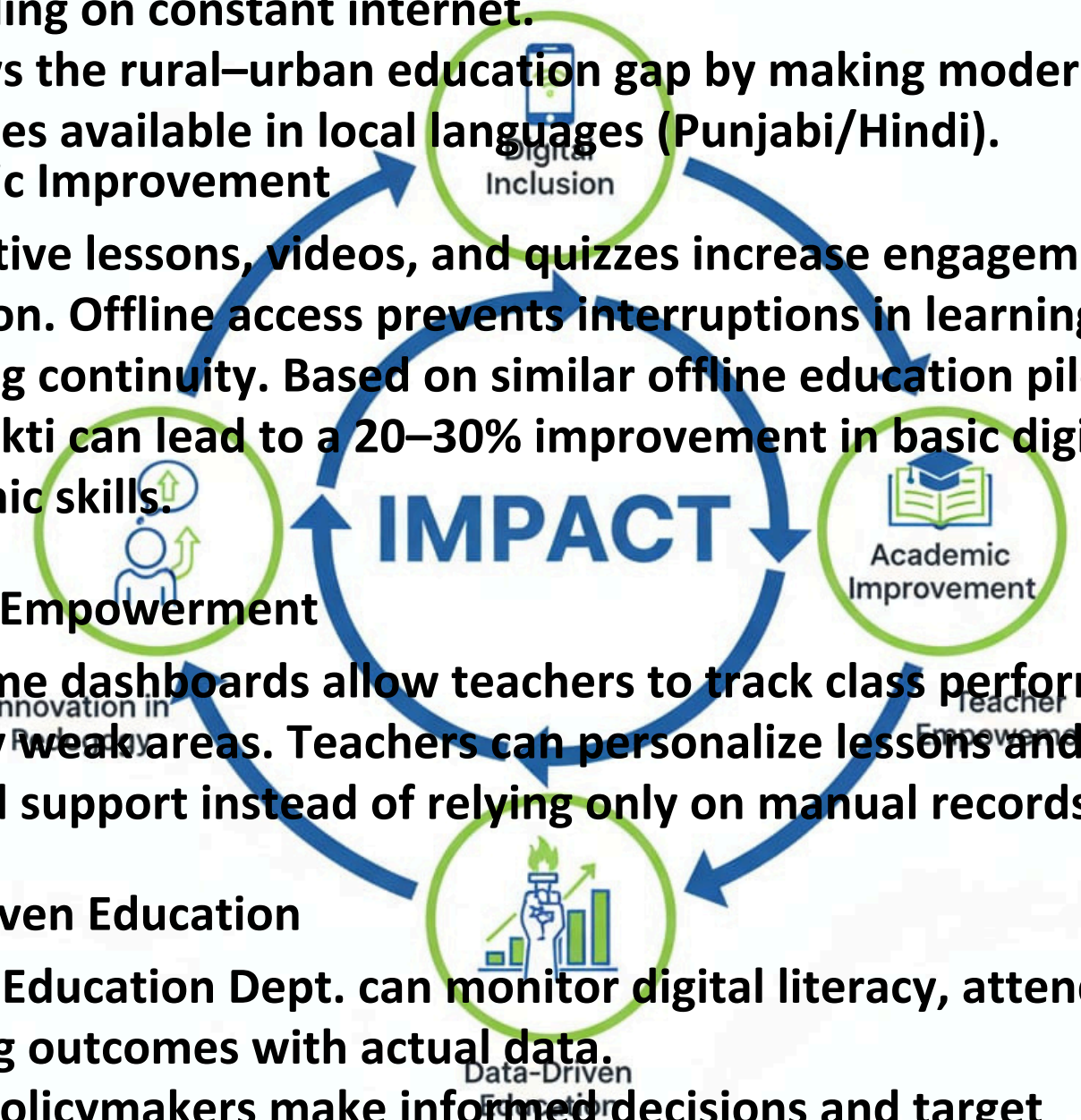
Interactive lessons, videos, and quizzes increase engagement and retention. Offline access prevents interruptions in learning, ensuring continuity. Based on similar offline education pilots, DigiShakti can lead to a 20–30% improvement in basic digital and academic skills.

Teacher Empowerment

Real-time dashboards allow teachers to track class performance and identify weak areas. Teachers can personalize lessons and give focused support instead of relying only on manual records.

Data-Driven Education

Punjab Education Dept. can monitor digital literacy, attendance, and learning outcomes with actual data. Helps policymakers make informed decisions and target interventions more effectively.



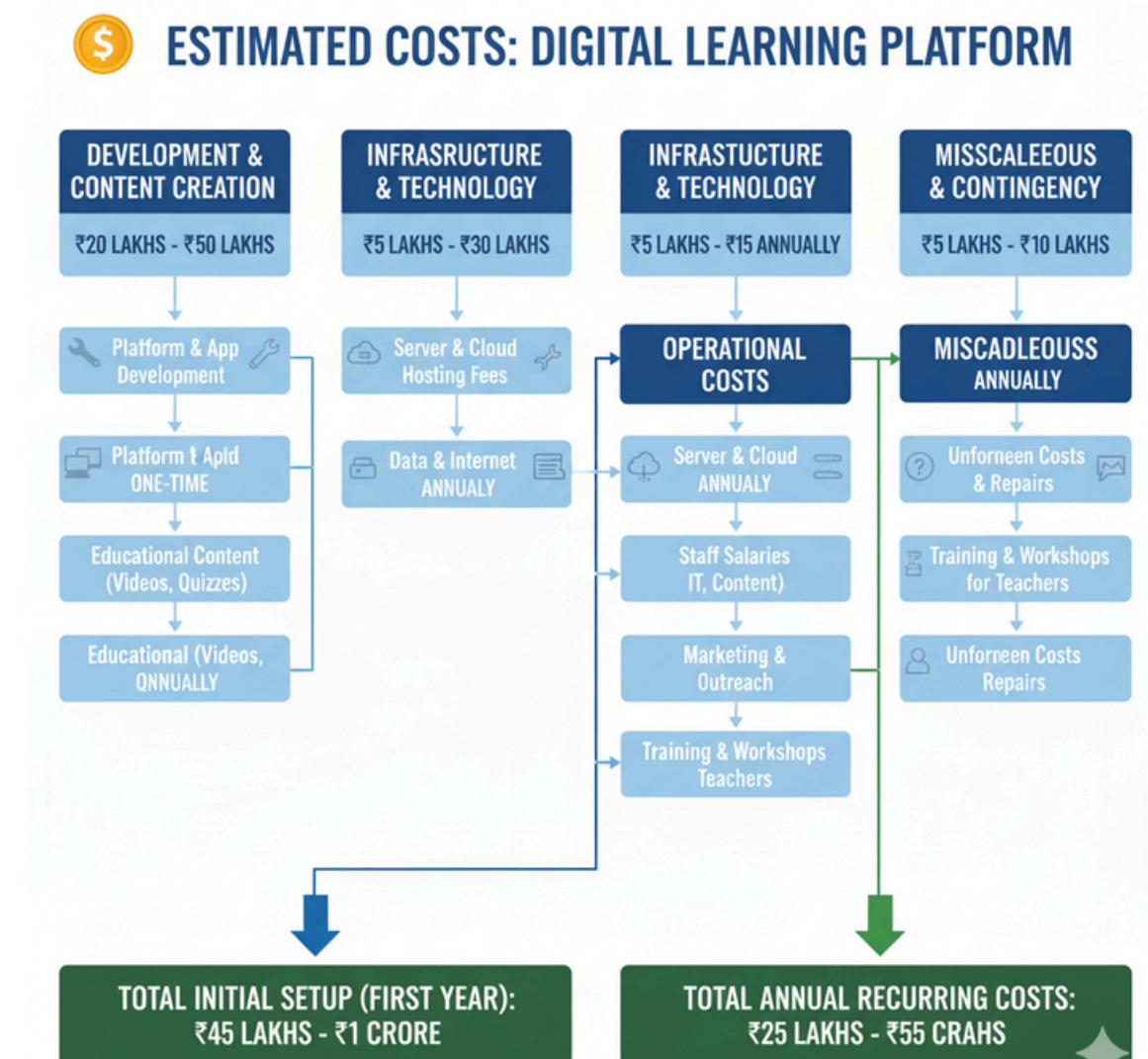
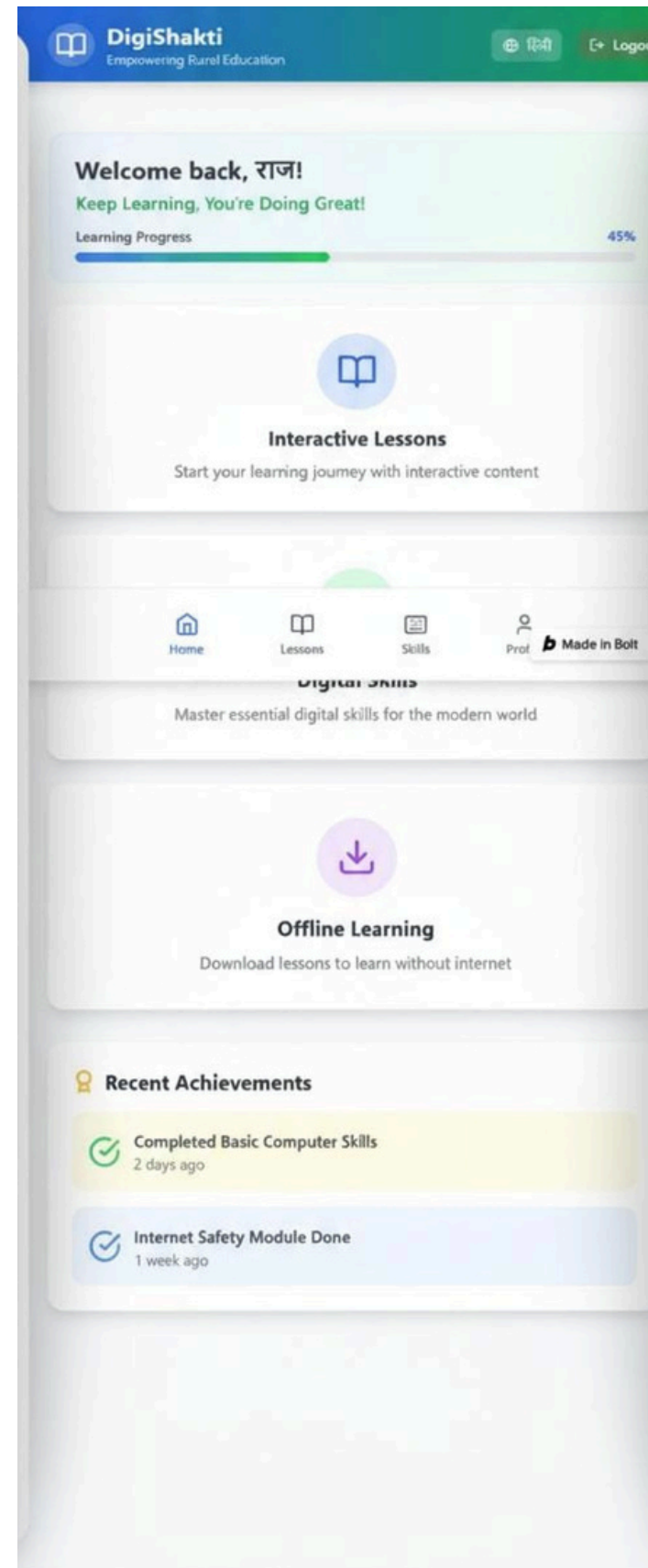
Prototype

Policy & Government

- National Education Policy (NEP 2020): Emphasizes digital literacy, local language content, and equitable access across rural and urban areas.
- Digital India is a Government of India initiative to transform the country into a digitally empowered society and knowledge economy.
- Punjab Govt. Initiatives: Digital classrooms, smart schools, and online content libraries already being rolled out in the state. DigiShakti can build on these.

Reports & Research

- Tribune India (2023) – Over 85% of rural students in Punjab have access to smartphones, but lack quality digital learning platforms.
- IAMAI (2022–23 Reports) – Rural India has seen 70% growth in mobile internet usage, yet digital literacy levels remain low.
- UNESCO & UNICEF Studies – Highlight a widening digital divide where rural students lack equal access to online learning.
- World Bank (2022) – “Digital Skills for Inclusive Growth in India” → stresses need for scalable, low-cost solutions for rural education.



Live Demo - <https://digishakti-rural-dig-c3lk.bolt.host/>